

Md Monirul Islam

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Education

- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering March 2018 – May 2023
CGPA: 3.85/4.00 (**3.98/4.00 in the final year**), ranked **25 of 189**, awarded with Honours
Relevant Courses: Electrical & Electronic Circuits, Calculus I & II, Ordinary & Partial Differential Equations, Linear Algebra, Communication System, Digital Signal Processing, Digital Electronics, Power Electronics, Control Systems, Microprocessors and Embedded Systems, Analog Integrated Circuits and Design, VLSI Circuits and Design

Research Interests

- Robotics • Autonomous Systems • Multi-Robot Coordination • Motion Planning • Path Planning • Real-Time Control • Sensor Fusion • Embedded Systems • Robot Perception • Human-Robot Interaction • Field Deployment of Robotic Platforms

Research Experience

- **Production multi-AGV fleet — centralized job management + decentralized path coordination.** Designed *FleetCore* (fail-closed ROS 2 coordinator: FIFO job dispatch, per-robot sequence-counter FSMs, mutual exclusion on shared resources, layered timeouts, single-button operator failsafe) paired with a fully decentralized A* planner — each robot publishes timestamped grid waypoints and peers treat them as spatiotemporal reservations (three conflict checks, continuous timestamp maintenance, plan-preserving obstacle handling). Months of zero-collision production operation on a 3-robot fleet in a live human-occupied factory.
- **Bilateral cross-coupled BLDC speed synchronization on a dual-MCU architecture.** Two BLDC motors each regulated by their own MCU (PID + velocity / acceleration feedforward); a bilateral cross-coupled term, evaluated identically on both nodes from a dedicated inter-MCU speed exchange, drives the inter-motor speed difference toward zero. Gains GA-tuned offline against an identified motor model; outperforms an independent-PID baseline on transient and steady-state synchronization.
- **Closed-form analytical IK with FK-anchored goal accumulation for a tracked mobile manipulator.** Tele-op platform combining a 6-DOF arm, a 2-DOF anti-flipper arm (also a collision-avoidance element), and a differential-drive base. From-scratch analytical solver replaces MoveIt's KDL with deterministic, branch-stable solutions; smooth turret-scaling preserves continuity through the home-pose singularity and an FK-anchored accumulator substantially reduces un-commanded cross-axis motion. MoveIt 2 handles motion-planned special commands (return-to-rest, named configurations) and per-tick collision validity across arm, anti-flipper, and carrier. Hardware assembly underway; once field-tested, set for UN peacekeeping deployment.
- **Latency-compensated 5-state EKF for visual-fiducial AGV localization.** Fuses wheel-encoder odometry with AprilTag camera fixes; velocity-bias states absorb systematic encoder error over long corridors; rewind-and-replay on a bounded encoder buffer applies each camera fix at its true measurement timestamp. Two operating modes (Kalman while moving, complementary while parked) hand off seamlessly across arrival / departure.
- **Fault-tolerant dual-network feedback architecture for unmanned multi-node robotic platforms.** Two redundant feedback networks carry coordination data between control nodes in parallel; the receiver detects channel degradation and falls back to the healthy network without losing a control cycle. Targets single-point-of-failure risks that surface only in field deployment, where any single wired or RF link can drop intermittently.
- **High-payload multirotor drone development.** Paper-stage design and aerodynamic simulation of a heavy-lift multirotor; built an in-house sizing optimizer that, given payload, flight time, motor performance data, motor count, and airframe parameters, computes feasible battery-pack configurations (hover-throttle reachability, per-motor current limits, voltage compatibility, cruise margin). Optimizer published as an interactive client-side companion artifact on the portfolio site.
- **α -Graphyne ballistic nanoribbon FET.** Undergraduate research with Dr. Mahbub Alam modelling quantum transport in 2D-material field-effect transistors.

Publications

- **Md Monirul Islam**, Mainul Islam, Emon Hasan Saumik, “Peer-Aware Distributed Path Planning for a Production-Deployed Multi-AGV Fleet in a Human-Occupied Garment Factory”, In Preparation, 2026 (first / corresponding author).
- **Md Monirul Islam**, Md Shakil Tanvir, Mainul Islam, Fiaz Ahmed Tonmoy, “Speed Synchronization of a Dual-BLDC Tracked Differential-Drive Vehicle Using a GA-Tuned Bilateral Cross-Coupled Controller”, Under Review, 2026 (first / corresponding author).
- Md Shakil Tanvir, **Md Monirul Islam**, Fiaz Ahmed Tonmoy, Mainul Islam, “A Dual-Network Feedback Communication System for Multi-Node Unmanned Robotic Control”, Under Review, 2026 (co-author).

- Md Shakil Tanvir, Fiaz Ahmed Tonmoy, **Md Monirul Islam**, Mainul Islam, Mahamudul Hasan, Raihan Ur Rashid Razeen, “Optimizing Master-Slave Broadcast Communication in Multi-Node Networks Using RS485 Standard”, 2024, 27th International Conference on Computer and Information Technology (ICCIT), IEEE. doi: [10.1109/ICCIT64611.2024.11022430](https://doi.org/10.1109/ICCIT64611.2024.11022430)
- H. Khan, **Md Monirul Islam**, R.I. Roya, S.N. Azad, “Performance Analysis of an α -Graphyne Nano-Field Effect Transistor”, *Micromachines*, MDPI, 2023. doi: [10.3390/mi14071385](https://doi.org/10.3390/mi14071385)

Professional Experience

- **Research and Development Unit, Cybernetics Hi-Tech Solutions (Pvt) Ltd** Dhaka, Bangladesh
R&D Engineer and Team Lead (June 2023 – present)
 - Lead the R&D engineering team taking robotic platforms from concept through field deployment; own architecture, coding, hardware integration, commissioning, and post-deployment support.
 - **CyberFleet (production AGV fleet)**. Lead engineer end-to-end. The platform productised the work and is now operating continuously at The Urmi Group RMG facility, Dhaka.
 - **Jontro Soinik v1 / v2 EOD ROVs**. Owned the electrical / firmware side across both generations. v1 was gifted to the Peruvian Armed Forces by the Bangladesh Army and is deployed under the UN Peacekeeping Mission in Mali (2024); v2 was Army-validated (disrupter and IP55 tests) and is set for Congo (2025). Ran on-site operator-training sessions and field hand-over to receiving units.
 - **Jontro Soinik 2.0 (6-DOF mobile manipulator)**. Owned the control / software stack for the tracked tele-op platform.
 - **Heavy-lift multirotor (50 kg payload)**. Owned BMS, propulsion sizing, frame, and power-distribution PCBs; shipped the sizing study as an interactive in-browser tool on the portfolio site.
 - **SD15-N target-tracking surveillance drone**. Designed the autonomous-tracking pipeline (vision → MAVLink → Pixhawk).
 - **Titan 550 industrial CoreXY 3D printer**. Two-iteration product build; the in-house fab loop that produces fixtures, brackets, and prototypes for every other platform on this list.
 - **In-house PCB & CNC**. Designed all controller / communication boards across the ROV & AGV platforms in Altium / KiCad and milled them in-house on a 3-axis CNC — no external fab.
 - **Other**: hybrid VTOL quad-plane prototype; real-time pet-food sorting line for an industrial-automation client.
- **Ulkasemi Pvt. Limited** Dhaka, Bangladesh
Intern (Dec 2022 – Mar 2023)
 - Worked on IC design using TSMC 180 nm and GPDK 90 nm technologies.
 - Developed a Wishbone interface and validated a high-speed BiFET Op-Amp in Verilog.
- **Walton Hi-Tech Industries Limited** Dhaka, Bangladesh
Industrial Trainee (Nov 2022)
 - Observed and analyzed PCB manufacturing and EV assembly workflows.
 - Gained exposure to industrial electronics production workflows.

Selected Undergraduate Projects

- **Real-time traffic speed estimation with YOLOv2**: Trained YOLOv2 in MATLAB to detect vehicles and estimate per-vehicle speed from monocular highway video.
- **5-DOF robotic arm (analytical IK)**: Designed in SOLIDWORKS, 3D-printed entirely on a self-built hobby printer using custom 3D-printed planetary gearboxes; closed-form analytical inverse-kinematics control on stepper motors.
- **Autonomous GPS-waypoint delivery drone**: Mechanical design in SOLIDWORKS, Arduino flight controller with PID stabilisation, GPS-waypoint navigation, and a custom payload-deployment mechanism.
- **Self-built hobby 3D printer**: Designed every component in SOLIDWORKS, fabricated the frame on a laser cutter, Arduino Mega controller running custom Marlin firmware. The workhorse that produced every other undergraduate project’s parts.
- **RV32I multicycle RISC-V processor + Wishbone memory controller**: Custom Verilog implementation with full testbench verification; physical-layout exploration in Cadence at TSMC 180 nm.
- **OMR machine from discrete digital logic**: Optical-mark recognition built entirely from discrete NAND / RAM / MUX / Comparator / LDR components — no microcontroller in the loop.
- **Regulated boost converter (TSMC 180 nm, Cadence)**: Schematic capture and physical layout of a voltage-regulated DC-DC boost converter.
- **Closed-loop induction-motor speed control**: V/f scalar control with SVPWM inverter modulation in MATLAB / Simulink.
- **Particle-concentration light-scattering sensor**: Laser + photodetector light-scattering setup with MATLAB-side calibration curves and signal-conditioning circuit.

- **Real-time environmental monitoring:** Solar-powered IoT node logging temperature, humidity, and CO₂ over GSM; Arduino + Autodesk Eagle PCB.
- **Laboratory variable DC power supply:** Linear-regulator design with current limiting and short-circuit protection; MATLAB efficiency study.
- **Electrical-services design (seven-storey building):** End-to-end electrical design: load estimation, panel scheduling, cable sizing, lighting / power layouts, riser diagram, earthing scheme.

Skills Summary

- **Robotics & Middleware:** ROS 2 (Jazzy), MoveIt 2, ros2_control, JointTrajectoryController, URDF / Xacro / SRDF, AprilTag, MAVLink, dronekit, pymavlink
- **Embedded & MCU:** STM32 (HAL & LL), ESP32, AVR, Arduino, ARM Cortex-M, µVision Keil, BigTreeTech 32-bit boards
- **Control & Algorithms:** Closed-form analytical IK, Extended Kalman Filter, PID (cascaded, GA-tuned), S-curve motion profiling, distributed A* path planning, peer-aware spatiotemporal reservations
- **Flight Control & Vision:** ArduPilot, iNav, Betaflight, Pixhawk, YOLOv11n, OpenCV
- **Simulation & Design Software:** MATLAB / Simulink, Cadence Virtuoso, PSPICE, PSAF, Icarus Verilog, GTKWave
- **PCB & Circuit Design:** Altium Designer, KiCad, Autodesk Eagle, Proteus; in-house 3-axis CNC milling for prototype boards
- **CAD & Mechanical Design:** SolidWorks
- **3D Printing & CNC:** Marlin (customised), Ultimaker Cura, Creality CR-6 SE; FlatCAM, Universal G-code Sender, GRBL
- **Programming Languages:** Python, C, Embedded C, Verilog
- **Documentation:** Microsoft Office Suite, L^AT_EX, Jekyll (al-folio)
- **Soft Skills:** Leadership, Project Management, Field Commissioning, Operator Training, Technical Writing

Honors and Awards

- University Merit Scholarship, BUET (three times — January 2019, January 2020, January 2022)
 - Awarded to students with outstanding academic record (GPA 3.90+) in a semester
- Dean's List Award, BUET (four times — all academic years)
 - Awarded to students obtaining an average GPA of 3.75+ in two consecutive semesters of an academic year
- University Stipend (three times — January 2018, July 2018, July 2021)
- Higher Secondary Certificate (HSC-2017) Board Scholarship
- Runners-up, Bangabandhu Creative Talent Hunt 2014, Pabna Division

Extra-curricular Activities

- Affiliate Member, BUET Robotics Society (BRS). (Jan 2022 – Mar 2023)
- Affiliate Member, Team Interplanetar – BUET Mars Rover Team. (Jun 2022 – Mar 2023)

Field Deployments

- 3-robot AGV fleet – live production at The Urmi Group RMG facility, Dhaka, 2025 – present
- Jontro Soinik v1 EOD ROV – UN Peacekeeping Mission, Mali, 2024 (Bangladesh Army → Peruvian Armed Forces handover)
- Jontro Soinik v2 EOD ROV – tested by Bangladesh Army, set for UN Peacekeeping deployment in the Republic of Congo, 2025

References

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